Biochem. Cell. Arch. Vol. 17, No. 2, pp. 757-760, 2017 | www.connectjournals.com/bca | *ISSN 0972-5075*

MANAGEMENT OF FOOT ROT DISEASE OF BLACK PEPPER THROUGH INTEGRATED APPROACHES IN NORTHERN HILL ZONE OF KARNATAKA

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(Accepted 21 October 2017)

ABSTRACT : The investigations on management of foot rot of black pepper was taken with different components of management options for two growing seasons. Among five different components, the application of Neem cake (2kg/vine) +*Trichoderma* 50 g / vine- Metalaxyl – MZ (1.25 gm / lit) *Pseudomonas fluorescens* spray (10 ml / lit), -Spray with Potassium phosphonate (0.3 %) has considerably reduced the foot rot incidence to the extent of 10.12 percent which is followed by Neem cake + Trichoderma-spary with Bordeaux mixture 1% (June)- Metalaxyl MZ (1.25g/l) July (18.35%), Metalaxyl – MZ (1.25 gm / lit (drench and spray)-June, July-Potassium phosphonate (0.3 % August) (23.41%) and Spray with Bordeaux mixture (1%) June Copper oxychloride drenching (July & August) (26.25%). However, the untreated control has recorded maximum foot rot incidence of 38.14 per cent. The gross income of Rs 1, 82,340/- was obtained in treatment the application of Neem cake (2kg/vine) +*Trichoderma* 50 g / vine-Metalaxyl – MZ (1.25 gm / lit) *Pseudomonas fluorescens* spray (10 ml/ lit) with the maximum net returns of Rs.1,69,140/ followed by Neem cake + Trichoderma-spary with Bordeaux mixture 1% (June)- Metalaxyl MZ (1.25g/ l) July which has recorded gross income and net returns of Rs. 1,23, 120/- and 1,05, 120/ respectively.Thus,integrated components can be recommended for the management of foot rot of black pepper.

Key words : Integrated components, foot rot, black pepper, Northern hill zone.

INTRODUCTION

Black pepper (*Piper nigrum* L.) the king of spices which earns better foreign exchange in India.In India, black pepper is being cultivated in Kerala (96%), Karnataka (3%) and to a lesser extent, in Maharashtra, Andhra Pradesh, Tamil Nadu and north eastern regions (Anonymous, 2005). India produces around 35% of the total world production of around 65000 t in the country about 70% is exported annually. In Karnataka, black pepper is cultivated in an area of 10,690 ha with a production of 2360 tonnes (Anonymous, 2005). It is mainly cultivated in Kodagu, Uttara Kannada, Dakshina Kannada, Shimoga and Chikmagalore districts as mixed crop in plantation such as arecanut, cardamom, coffee and rubber whereas, in Kerala it is being grown as pure crop trained on *Erythrina indica* Lam. K.

The major production constrains in the black pepper cultivation have been attributed mainly for pronounced mortality of vines by the dreaded disease foot rot caused by *Phytophthora capsici* emend, Alizedeh and Tsao, old gardens occupied with traditional cultivars having poor genetic potential, non-adoption of improved package of practices and improper management of gardens. In Uttara Kannada, which is situated in upper region of Western Ghats, black pepper is mainly cultivated in moist valleys in arecanut plantations wherein arecanut trees serve as live standards for training black pepper. Due to epiphytotic appearance of *Phytophthora* foot rot disease of black pepper most of the gardens have been wiped out since 1978 (Sastry, 1982 and Dutta, 1984). The *Phytophthora* foot rot disease occurs mainly during the South-West monsoon period (June to September) where, high soil moisture, high relative humidity, shorter duration of sunshine hours, high and well distributed rainfall and low temperature exists and conducive for the disease development (Sarma *et al*, 1988). All parts of the vines are affected and the symptoms expression depends upon the site or plant parts infected and the extent of damage (Anannadraj *et al*, 1988).

Earlier workers recommended one per cent Bordeaux mixture to manage the disease as preventive fungicide. But after the onset of the disease the fungicide does not give satisfactory control of root and collar infection. Integration of chemicals and biological agents for managing soil borne disease has been considered as a novel approach, as it requires low amounts of chemicals, thereby reducing cost of control as well as pollution hazards, with minimal interference of biological equilibrium (Papavizas, 1973). Therefore, an attempt has been made by krishi Vigyana Kendra, Uttar Kannada, Sirsi during

2007-08 and 2008-09 to integrate chemicals, biological agents and organic amendments to manage the disease in the farmer's fields.

MATERIALS AND METHODS

A field experiment on management of foot rot of black pepper by using chemicals, biological agents and plant product like neem cake was carried out in a farmer's garden at Navanagere village of Sirsi taluk of Norhern hillzone of Karnataka during 2007-08 & 2008-09. Integrated disease management of foot rot of black pepper was carried using Randomized Complete Block Design (RCBD). The treatment details are given below:

Treatment details

Treatments	Details
T1	Spray with Bordeaux mixture (1%) June Copper oxychloride drenching (July & August)
T2	Metalaxyl – MZ (1.25 gm/lit (drench and spray)-June, July-Potassium phosphonate (0.3% August)
T3	Neem Cake(2kg/vine)+Trichoderma 50 g /vine 15 days before monsoon-Metalaxyl – MZ (1.25 gm/lit) (drench and spray)-June-Pseudomonas fluorescens spray (10 ml / lit) - (July, August)-Spray-Potassium phosphonate (0.3 % August)
T4	Neem cake + Trichoderma-spary with Bordeaux mixture 1% (June)- Metalaxyl MZ (1.25g/l) July
T5	Un treated control (without treatment)

The spray schedule was initiated as per the schedule irrespective of disease incidence.

In each treatment, 20 pepper vines were selected. The treatments were imposed as prophylactic and during the monsoon.

Application of neem cake

Finely powdered neem cake was incorporated into top 10 cm layer of soil around the vines at 2 kg/vine.

Application of bio control agents

Trichoderma harzianum was mass multiplied on moist wheat bran preparations (1:1 v/v) (Jahagirdar, 1998). It was added to soil at rate of 50 g of preparation per vine. *Pseudomonas fluorescens* was mass multiplied on King's B broth (Sivamani and Gnanamanicum, 1988) (and the cell count was adjusted to 108 cells/ml) and the biocontrol agent was diluted in 3 l of water and then applied to the vines.

Application of botanicals & fungicides

The botanicals & fungicides , viz. Potassium phosphonate (0.3%), Metalaxyl MZ 72 WP (Ridomil MZ 72 WP) (0.125%), Bordeaux mixture (1%) and Copper oxychloride (0.2%) were applied as drench and spray along with biocontrol agents either alone or in combination.

Observations

Foot rot infection

Collar infection was recorded and presented as per cent collar infected vines using the formula :

$$\frac{\text{Number of foot rot infected vines}}{\text{Total number of vines}} \times 100$$

Leaf infection

Number of vines showing leaf infection was recorded and presented as per cent leaf infection. For intensity of leaf infection, three areas (0.5 sq. m.) randomly selected in the canopy of black pepper vines, preferably each at lower level, middle level and upper level of the canopy and number of leaves present and number of leaves infected by the disease were recorded and presented as per cent leaves infested by the disease (Anonymous, 2004).

Yields and economics

Yields of fresh berries/vine was recorded by harvesting separately and converted it in to q/ac. The economics like cost of cultivation, gross returns and net returns was calculated as per the prevailing prices in the Sirsi market.

RESULTS AND DISCUSSION

The results from the above table revealed that, integration of organics; manures, botanicals and inorganic fungicides have greater influence in reducing the disease incidence, increasing the yields and maximum returns. The application of Neem cake (2kg/vine) + *Trichoderma* 50 g / vine-Metalaxyl – MZ (1.25 gm / lit) *Pseudomonas fluorescens* spray (10 ml / lit), spray with Potassium phosphonate (0.3 %) has considerably reduced the foot rot incidence to the extent of 10.12 percent (T3). This is followed by T4 (18.35%), T2 (23.41%) and T1 (26.25%). However, the untreated plots (T5) has recorded maximum foot rot incidence of 38.14%. The leaf infection was maximum in unsprayed plots (26.25%) and minimum in T3 (5.34%), which is significantly superior followed by T4 (12.23%). The proportionately higher yields of 10.13 q/ac fresh berries were recorded in the best integrated treatment (T3) followed by T4 (6.84 q/ac), T2 (2.68q/ac) & T1 (2.15q/ac). The gross income of Rs 1, 82,340/- was obtained in treatment T3 with the maximum net returns of Rs. 1,69,140/ followed by T4 which has recorded gross income and net returns of Rs. 1,23,120/- and 1,05,120/ respectively. Integrated management is the harmonious blending of various methods in proper sequence and timing so as to create least harmful effects on man and environment. The various components of integrated management are eco-friendly, economically

Table 1 : Integrated management of foot rot of black pepper.

Treatments	Details	Foot rot incidence (%)	Leaf infection (%)	Yield of dry berries (q/ac)	Gross income Rs/ha	Net income Rs/ha
T1	Spray with Bordeaux mixture (1%) (June) and drenching with Copper oxychloride (July & August)	26.25	18.41	2.15	38,700	25900
T2	Metalaxyl – MZ (1.25 gm/lit (drench and spray)-June, July. Spray with Potassium phospho-nate (0.3 % August)	23.41	16.35	2.68	48,240	41,840
T3	Neem cake (2kg/vine)+*Trichode-rma* 50 g /vine 15 days before monsoon-Metalaxyl–MZ (1.25gm / lit) (drench and spray)-June-*Pseudomonas fluorescens* spray (10 ml/lit) - (July, August)-Spray-Potassium phosphonate (0.3% August)	10.12	5.34	10.13	1,82,340	1,69,140
T4	Neem cake + Trichoderma-spary with Bordeaux mixture 1% (June)-Metalaxyl MZ (1.25g/l) July	18.35	12.23	6.84	1,23,120	1,05,120
T5	Untreated control	38.14	26.25	0.96	17280	
	CD @ 5 %	9.37	5.35	0.52		
	CV	21.22	18.18	10.09		

Market rate of black pepper @ Rs 18,000/q.

feasible and compatible. Conceptually, Integrated Disease Management (IDM) or Integrated Pest Management (IPM) involves the selection, integration and implementations of control based on predicted economic, ecological and sociological constraints. This technology therefore, seeks maximum use of naturally occurring controls including weather, disease agents, native as well as introduced antagonists. In the present investigation combination of Neem Cake(2kg/vine)+*Trichoderma* 50 g/vine 15 days before monsoon-Metalaxyl–MZ (1.25 gm/ lit) (drench and spray)-June-*Pseudomonas fluorescens* spray (10 ml/lit) - (July, August)-Spray-Potassium Phosphonate (0.3% August) has found to be significant treatment in minimizing the disease incidence and maximizing the yields and net income. These results are in agreement with the following studies. Two isolates of *Pseudomonas corrugata* (PC 13 and PC 35) and three of Pseudomonas fluorescens (Pf15, Pf16 and Pf 27) were used in the biological control of *Pythium* root rot of cucumber Pf15 isolate significantly reduced the disease incidence and produced 85% more marketable fruits than rest of the treatments (Rankin and Paulitz, 1994).

Use of biocontrol agents like *Glomus fasciculatum*, *T. hamatum* and *G. virens* were found effective against *Phytophthora* foot rot of pepper (Anandaraj *et al*, 1995). Biocontrol studies against foot rot disease revealed that *T. viride* was found to be effective in controlling wilt disease with incidence (10%). In biocontrol studies of *Phytophthora* foot rot disease, use of antagonistic organism *T. viride* @ 150 ml/vine along with 5 kg of FYM was found effective in trials at Sirsi and Panniyur (Anonymous, 1997).

Field experiments have shown that, incorporation of biological agents in the field suppresses the soil borne inoculum (Anandarj and Sarma, 1994). Various formulations of metalaxyl and phosphorous acid are also found effective under experimental conditions (Ramachandran *et al*, 1991). Studies with newer fungicides revealed that the treatment consisting of all cultural practices + 1 kg neem cake + 3 g phorate/vine + first spray with 1% Bordeaux mixture + 0.2% Potassium phopsonate (second) recorded minimum defoliation (6.74%) and foliar yellowing (5.65%) without any death of vines (Anonymous, 1997). Spraying with 1% Bordeaux mixture + drenching with 0.2% copper oxychloride was found effective against *Phytophthora* foot rot of black pepper (Anonymous, 1996). Studies at Panniyur including standardization of management practices with spraying of 0.2% Akomin followed by spraying and drenching with 1% Bordeaux mixture were found to be effective in checking foot rot disease under different shaded conditions (Anonymous, 1997). Jahagirdar *et al* (2000) recorded lower disease incidence with the treatments *Trichoderma viride* + Metalaxyl + Akomin; Metalaxyl + Akomin;

Metalaxyl alone; *T. viride* +MPG-3. The per cent increase in yield was above 50 per cent in all treatments, compared to the control. Bio control agent *Trichoderma harzianum* along with potassium phosphonate has recorded highest disease suppression with least foliar yellowing (Srinivasan *et al*, 2003).

Hence, integrated treatment combination of soil manures, bioagents, botanicals, prophylactic and curative sprays of fungicides could be used to manage the foot rot and leaf infection caused by *Phytophthora capcisi* in high rainfall regions of North hill zones Karnataka

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